

When Modal Contact Rows Fail in Fixed-Budget XPBD: Diagnosis and an Operating Guide

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Abstract

Rigid-body XPBD engines already resolve contacts and joints, and adding a small global modal state (16–24 amplitudes) is an attractive way to give a stiff prop low-cost vibration without converting it into a full nodal/tet deformable. We show that inserting this globally shared modal state directly into a two-way unilateral contact row is not automatically energy-safe when the host runs only a few local iterations per substep. Across three scenes, two relaxations and four schedules, *our tested XPBD implementation* overdraws the measured gross rigid-side supply by up to 4.4×10^7 J (a $2.2\text{--}4.5 \times 10^7$ J neighborhood over a deterministic perturbation ensemble), while augmented-Lagrangian and implicit implementations are markedly milder controls (≤ 6.7 J in 3 of 24 cells and never, respectively; a gravity/no-contact accounting audit places the control floor below 10^{-3} J, so 6.7 J is real injection). A same-path iteration ladder, equal-row-count schedules, warm-starting and mode truncation localize the catastrophic XPBD amplification to finite-budget convergence of the shared modal coupling, and an in-host block-condensation ablation shows condensing the shared rows does *not* fix it. When the contact architecture is flexible the tested stiffness-aware implicit realization is the preferred observed control; in a fixed XPBD host, spend the local budget on iterations and verify the modal basis — more substeps and a block solve do not suffice. For hosts that still cannot converge, a cumulative modal-storage projection enforces its scalar accounting bound unconditionally (Prop. 2.1, roundoff floor in all 90 measured cells) but opens up to 21.6 mm of penetration (72% of the board’s thickness), and is therefore containment, not corrected contact.

CCS Concepts

• Computing methodologies → Physical simulation.

Keywords

contact, modal reduction, position-based dynamics, energy safety, fixed-budget solvers

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1 Introduction

A developer building an interactive scene on a rigid-body XPBD engine [11] already has a solver that resolves contacts and joints under a small, fixed local-iteration budget. To make a stiff prop — a shelf, a table, a ledge — ring when it is struck, the tempting move is not to replace it with a full nodal or tetrahedral deformable, which adds thousands of online coordinates and local internal constraints,

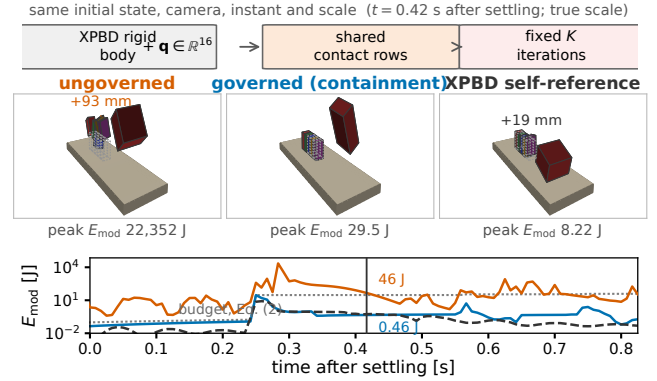


Figure 1: One shelf impact at a production-like interactive budget (1×8), true scale, three independent runs. Ungoverned, truncation energy throws the resting books up to 93 mm — more than their own height; governed, (2) holds peak modal energy to 29.5 J and they stay within 1 mm of rest; the reference is the host’s own high-iteration self-reference (500×1). The bound removes the spurious launch and some legitimate motion with it: the reference lifts one book 19 mm, the governed run none.

but to attach a small precomputed modal basis: 16–24 global amplitudes $\mathbf{q}$ capturing the low-frequency bending of a stiff, small-displacement body [8, 9, 13]. The runtime state grows by a handful of numbers, the precomputation is offline, and the same XPBD host steps everything. This paper is written for that developer, not to argue XPBD against another solver.

The catch is in the contact. A modal prop driven *one way* — an external filter reads detected contact events and rings the modes with no back-reaction — is the real-time precedent [3], but it double-counts the interface energy and cannot let the prop’s own deformation change how it is supported. The physically consistent move is a *two-way* row: rigid and modal unknowns resolved together through one shared contact multiplier [17], so a heavier load bends the support and the bent support redistributes the load. The trap is that every support contact touches the *same* global vector $\mathbf{q}$, so the contact-row graph is dense on the modes even though the state is tiny — and a host running only a few local iterations per substep need not converge that shared coupling. We take the row as given: at position level it is one linearization step from the velocity-level law ($\mathbf{C}^{n+1} \approx \mathbf{C}^n + h \mathbf{J} \mathbf{u}^{n+1}$, same Jacobian and multiplier as Zheng and James [17]), narrower than the original — normal rows only, $e=0$ restitution — and *not* a contribution of this paper. Our question is the practitioner’s: what does the obvious two-way transcription cost at a normal budget, and what should one do about it? *If solver*

architecture is unconstrained our evidence favors the tested stiffness-aware velocity-level realization; our target is the existing fixed-budget XPBD host.

Finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [15], and velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative solvers named, goes back at least to Kaufman et al. [10]. What is missing, and what we supply, is a quantitative account of this case — unilateral contact-to-modal transfer in a fixed-budget host — and an operating guide for the practitioner who hits it.

Contributions.

- (1) **A practitioner workflow and its failure.** We treat adding a compact modal state to a fixed-budget XPBD host as the workflow it is, and show that the direct two-way contact row overdraws its measured rigid-side supply catastrophically at interactive budgets — across three scenes, two relaxations and four schedules (§3.1) — with augmented-Lagrangian and implicit implementations as milder controls. The cross-control reading is secured by a gravity/no-contact accounting audit and the failure’s robustness by a deterministic perturbation ensemble.
- (2) **Mechanism and an operating guide.** A same-path iteration ladder, equal-row-count schedules, warm-starting, mode truncation and an in-host block-condensation ablation localize the amplification to finite-budget convergence of the shared modal coupling (§3.2), and yield an ordered decision rule: the tested implicit realization when the architecture is flexible, otherwise iterations and a verified basis — not more substeps, not a block solve — in a fixed XPBD host.
- (3) **A last-resort guardrail, priced.** A cumulative modal-storage bound funded by the measured rigid-side loss and enforced by a scalar projection of the realized state (§2): a deliberately minimal fail-safe, not a contact method, guaranteed unconditionally (Prop. 2.1, roundoff floor in all 90 measured cells) but at a measured cost to contact validity (§3.3).

Related work. Rigid XPBD [11], interactive modal deformation and velocity-level modal contact [8–10, 17], and reduced XPBD [13] set the stage; the last uses modes as a *deformation* surrogate, whereas here modal amplitudes enter as contact DOFs driven by the shared multiplier (cf. ray-traced penalty contact without a shared unilateral multiplier [18]). Our last-resort guardrail transplants energy-tank control [5, 7, 14] rather than inventing one: we keep the tank-and-observer structure but fund the tank from *measured* rigid-side loss — not the port it regulates — and scale realized *state* rather than a force, because a fixed-budget solver commits its multipliers before the excess is observable. This separates it from total-energy projection [4], bidirectional total-energy tracking [16], and the structural finite-budget passivity of *bilateral* couplings [15].

2 The row and the cumulative bound

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2} \dot{\mathbf{q}}^\top \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^\top \mathbf{K} \mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C = y_c - (y_{\text{rest}} + \mathbf{U}_y^\top \mathbf{q}), \quad C \geq 0, \lambda \geq 0, C\lambda = 0, \quad (1)$$

with $\partial C / \partial \mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law. It is one row *law*, instantiated at every support contact every substep — 48 such rows on the shelf, 40 on the ledge, 200 on the table — all sharing the one modal state $\mathbf{q}$.

The invariant. We require that the energy stored in the modal subsystem never exceed the *measured gross rigid-side loss* — defined exactly at (3) below — cumulatively over the run:

$$E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \eta \sum_{k \leq n} \max(\Delta E_{\text{rig}}^k, 0), \quad \eta \in (0, 1]. \quad (2)$$

The transfer efficiency η sets how much of the measured loss may be spent; *every experiment in this paper runs* $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is *cumulative*, not per-step: a substep may deposit more than it earns provided the running total remains within the budget, which is what lets a legitimate ring persist across substeps.

Definitions. Modal storage is the mechanical energy of the reduced state as written above, and $E_{\text{mod}}^0 = 0$: every scene starts from an undeformed, resting support, so even the run’s own settling transient must be funded by the reservoir. The supply is

$$\Delta E_{\text{rig}}^k = (E_{\text{rig}}^{k,-} - E_{\text{rig}}^{k,+}) + W_g^k, \quad W_g^k = \sum_b m_b \mathbf{g} (\mathbf{x}_b^{k,+} - \mathbf{x}_b^{k,-}), \quad (3)$$

where $E_{\text{rig}} = \sum_b [\frac{1}{2} m_b \|\mathbf{v}_b\|^2 + \frac{1}{2} \omega_b^\top \mathbf{I}_b \omega_b]$ is *purely kinetic* — translational and rotational, over all dynamic bodies, not merely those in contact — and $\pm$ denote the two ends of substep k . We call this the **measured gross rigid-side loss**: the scene-wide, gravity-corrected, gross kinetic-energy loss of the rigid subsystem. It is a *loss*, not a dissipation — it also counts legitimate rigid-to-modal transfer and losses at rigid-rigid contacts elsewhere in the scene — so it is an envelope on what contact dissipated, not a measurement of it (§4). Gravity enters only through W_g , so free ballistic motion registers zero supply — a falling body’s kinetic gain exactly equals the gravity work, and only an inelastic velocity change funds the reservoir. Substep-boundary endpoints tile the timeline exactly, so no rigid energy change escapes the accounting.

Two objects. (i) **The printed invariant** is (2) exactly as written: a cumulative storage ceiling on the modal subsystem whose supply is *measured, not prescribed* — the right-hand side is the measured loss of (3), never a tuned constant or a fraction of incident energy. It constrains a scalar reservoir, not a contact port, so it is neither port passivity nor a signed per-interface bound. The *gross* sum ignores substeps in which the rigid subsystem *gains* energy, and counts energy returned and re-dissipated as fresh supply (§4 measures both). (ii) **The enforced recursion** is the per-substep loop below — credit, test, project, debit — and maintains (2) exactly, not a relaxation: crediting the reservoir *before* the projection test puts the same-substep deposit that funds a legitimate rise already inside the

budget, so no granularity slack is required. Proposition 2.1 proves this; the only give is a per-substep accounting tolerance, realized at the 10^{-13} J roundoff floor.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage. Both happen *after* the contact solve, within the same substep: ΔE_{rig} is not observable until the velocity solve has finalized the rigid velocities, so the substep credits $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ at that point, tests the realized modal energy against it, and only then debits $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$. If the realized modal energy would overdraw B , the state is projected onto the admissible set by a single scalar,

$$(\mathbf{q}, \dot{\mathbf{q}}) \leftarrow \gamma(\mathbf{q}, \dot{\mathbf{q}}), \quad \gamma = \min\left(1, \sqrt{\frac{E_{\text{mod}}^- + B}{E_{\text{mod}}^+}}\right), \quad (4)$$

with $E_{\text{mod}}^\mp$ the modal energy before and after the solve. One substep of the governed loop, identical in all three hosts, then reads: (1) snapshot E_{rig}^- (pure KE, all bodies), $E_{\text{mod}}^-, \mathbf{x}^-$; (2) predict (gravity), generate contacts at the predicted pose; (3) position solve (K iterations), then velocity solve – the contact solve; (4) form $\Delta E_{\text{rig}} = (E_{\text{rig}}^- - E_{\text{rig}}^+) + W_g$, i.e. (3), post-solve; (5) credit $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$, same substep; (6) if $E_{\text{mod}}^+ > E_{\text{mod}}^- + B$, scale the state by γ of (4); (7) debit $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$ – contact is **not** re-solved.

The form (4) is closed because E_{mod} is quadratic in the state: scaling $\mathbf{q}$ and $\dot{\mathbf{q}}$ together scales both the kinetic and the elastic term by γ^2 , so γ bounds the total energy rather than the velocity alone. Two consequences are measured below: the projection is inert whenever the solve is within budget ($\gamma = 1$, bit-wise-identical trajectories), and it never re-solves contact (step 7).

PROPOSITION 2.1. *For any $\eta \in (0, 1]$ and any host, the loop maintains $B \geq 0$ and guarantees (2) unconditionally, assuming nothing about the contact solve, up to an accounting tolerance ϵ over n substeps (ϵ the per-test tolerance, 10^{-12} J).*

Proof. Write $c_k = \eta \max(\Delta E_{\text{rig}}^k, 0)$ for the substep credit and B_k for the reservoir after substep k , and carry two invariants: $B_k \geq 0$ and $(E_{\text{mod}}^k - E_{\text{mod}}^0) + B_k \leq \sum_{j \leq k} c_j$. Both hold at $k=0$ ($B_0 = E_{\text{mod}}^0 = 0$). The credit (step 5) enters B *before* the test, so step 6 applies the largest γ with $\gamma^2 E_{\text{mod}}^+ \leq E_{\text{mod}}^- + B_{k-1} + c_k$; as E_{mod} is quadratic in the state, realized storage is at most $E_{\text{mod}}^- + B_{k-1} + c_k$, i.e. the realized gain $\Delta E_{\text{mod}} \leq B_{k-1} + c_k$. The debit (step 7) sets $B_k = \max(B_{k-1} + c_k - \max(\Delta E_{\text{mod}}, 0), 0)$, keeping $B_k \geq 0$; and in either branch ($E_{\text{mod}}^k - E_{\text{mod}}^0) + B_k \leq (E_{\text{mod}}^{k-1} - E_{\text{mod}}^0) + B_{k-1} + c_k$, so the second invariant follows by induction. Since $B_n \geq 0$, this is $E_{\text{mod}}^n - E_{\text{mod}}^0 \leq \sum_{k \leq n} c_k$ – (2) itself, not a relaxation: the same-substep credit absorbs the granularity term a naive telescoping would otherwise leave. The 90 governed cells confirm it at the roundoff floor, worst margin 1.1×10^{-13} J.

3 Results

All solver-behaviour measurements below come from a single machine (Apple M4, CPU, CPython 3.12); every number is frozen with its generating command and commit hash in the supplemental ledger. We deliberately avoid mixing machines: chaotic contact stacks diverge across architectures under floating-point reassociation.

3.1 One row, three implementations, the same 24 cells

We drive an impactor into a modally reduced support in three scenes – a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [3] – and sweep modal relaxation and the local budget (iterations $\times$ substeps) over $\{4 \times 1, 8 \times 2, 16 \times 4, 32 \times 8\}$. The hosts are XPBD [11], AVBD [6] (a descendant of vertex block descent [2]), and a sequential-impulse realization in the sense of Catto [1], made implicit in the modal block. All three are pinned to $h=1/120$ s, per-substep contact regeneration, modal rank 16/16/24 (shelf/ledge/table), Rayleigh damping, an *implicit-midpoint* modal restoring step, and $\eta=1$; they differ in the unknown solved for (position / position + multiplier / velocity), the modal contact weight – XPBD’s explicit per-mode compliance $1/(H_{ii}h^2-1)$ versus the impulse host’s implicit $(\mathbf{M}+h\mathbf{D}+h^2\mathbf{K})^{-1}$ – warm start, and relaxation (the full host/parameter table is in the supplement). Two differences deserve emphasis. Relaxation is *inert* on the impulse host (its implicit modal weight needs no under-relaxation), so its two relaxation rows are bit-identical. Warm start also differs – $\lambda \leftarrow 0$ every substep for the position-based host, carried duals for the others – so the comparison is as-deployed, not compliance-matched. It is not the lever, though: warm-starting it leaves the footprint unchanged ($R > 1$ in the same 8/24 cells, worst 1.20×10^5 either way) and *worse* at 4×1 ($3.1 \times$), as a warm multiplier cannot converge the shared modal DOF that only iterations do (§3.2).

Equal row evaluations, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis spans $\times 64$ and conflates two refinements that are neither interchangeable nor equally priced (a substep also repeats contact generation and integration). At 32 row evaluations on the shelf drop, spent as iterations ($K=32, S=1$) the ratio is 0.300 and (2) holds; spent as substeps ($K=4, S=8$) it is 3.13, overdrawing by 481 J: substep refinement alone does not converge the row.

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1 : peak modal energy 4.44×10^7 J against an incident 371 J); the augmented-Lagrangian host in 2 of 24 – both at the starved 4×1 corner of the table scene, worst 1.70 – so it is not unconditionally benign either. The ordering is scene-dependent and it inverts: worst-over-scenes at 4×1 the position-based host is five orders of magnitude more energetic in the ratio R (1.2×10^5 vs 1.70), yet on the table scene itself the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

The ratio is not the invariant. An impactor’s incident kinetic energy is not the supply (2) budgets against, so we measure (2) directly on all three hosts, ungoverned, running the reservoir accounting live while forcing $\gamma \equiv 1$: every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an ungoverned run (the ratios above reproduce the independent sweep exactly). We report the signed margin of (2) in joules; positive values violate. The invariant and the ratio do not flag the same cells: the position-based host violates (2) in 9 of 24 cells against $R > 1$ in 8, the augmented-Lagrangian host in 3 against 2, and the implicit host in neither (0/24,

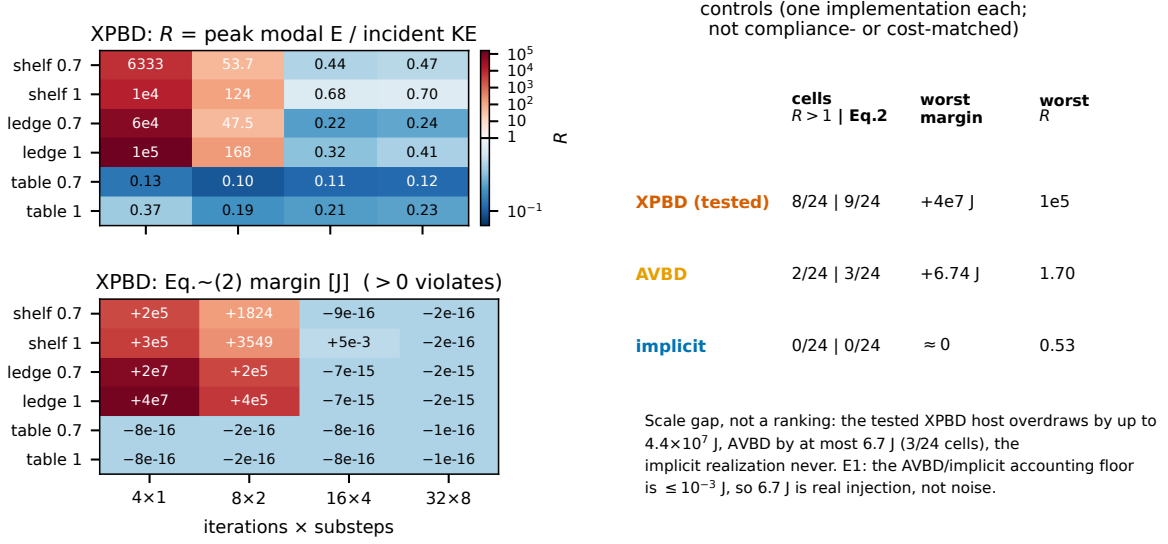


Figure 2: Our tested XPBD implementation, over the 24-cell sweep (3 scenes $\times$ relaxation $\{0.7, 1.0\} \times$ budget), governor OFF. Left, top: $R = \text{peak modal energy} / \text{peak incident rigid KE}$, a severity diagnostic; bottom: the signed margin of (2) in joules (> 0 violates). The two disagree by design (XPBD: $R > 1$ in 8/24, margin > 0 in 9/24) — a result, not noise. Right: a compact control strip for the three implementations (one of each, *not* compliance- or cost-matched): violating cells, worst margin, worst R . This is a scale gap, not a ranking — 4.4×10^7 J vs. ≤ 6.7 J vs. 0; the E1 accounting audit puts the control floor below 10^{-3} J, so 6.7 J is real injection. The full two-metric 72-cell three-implementation heatmap is in the supplement.

worst margin at the roundoff floor). Each metric catches a cell the other misses, so neither subsumes the other. What orders the failures is magnitude: the position-based overdraw reaches 4.4×10^7 J, seven orders past the largest augmented-Lagrangian one in energy (6.7 J, at the starved 4x1 table corner where $R = 1.70$), itself past the implicit host's exact zero — three regimes the ratio-to-incident-energy diagnostic compresses into a single number.

With enforcement enabled, all 72 cells of *this sweep* satisfy (2) — 60 of them distinct configurations, the implicit host's 12 relaxation pairs being bit-identical (Fig. 2); the 18 deployed-budget cells of §3.2 satisfy it too. The worst realized ratio falls to 1.22 (XPBD) and 0.87 (AVBD), and the projection never activates in any impulse cell, nor at 16x4 or 32x8 on the position-based host at relaxation 0.7: it is inert precisely where the solve is already within budget.

3.2 The position-based amplification is a truncation artifact

Figure 3 isolates the local iteration budget. Two references recur below and are not the same object: *the implicit reference* is the implicit high-iteration realization at $K=500$ — the same code path run to $K=500$ — and *the XPBD self-reference* is the position-based host run to $K=500$ against itself. The position-based host falls monotonically from 2.96×10^4 at $K=1$ to 0.300 at $K=32$, approaching the implicit reference 0.2735; the gap $|\text{XPBD} - \text{reference}|$ shrinks from 2.96×10^4 to 2.7×10^{-2} . The implicit realization moves only in the fourth decimal across $K = 2 \dots 500$ (max-min 6.8×10^{-4}).

A falling energy ratio does not by itself show that the *row* is converging — a single scalar could shrink coincidentally. Measuring

the constraint directly resolves the question, and the two complementarity conditions must be read in their own units rather than compounded into one dimensionally mixed scalar. End-of-substep penetration falls from 27.9 mm at $K=1$ to $3.6 \mu\text{m}$ by $K=64$, where it saturates. The multiplier surviving on *separated* rows — force where the gap is open, in units of the steady-state median — instead rises from $84\times$ to $139\times$ through $K=24$, collapsing to exactly zero only at $K=64$. The energy crossing at $K \approx 24$ therefore coincides with the *gap* converging, not with complementarity holding: the multiplier side clears a rung later. Position-based host only — the augmented-Lagrangian multiplier is not the same object, and the implicit realization is converged at $K=2$.

Self-convergence stops short of the reference. Running the position-based host to $K=500$ against itself does not close the gap: its ratio plateaus by $K \approx 64$ at 0.2996, so the residual 2.6×10^{-2} against the reference's 0.2735 is a fixed point, not a tail. The state agrees no better — on the deflection trajectory $d_i(t) = U_y[i]^T q(t)$ the peak matches to 2.4% but the trajectories differ by 6.8 mm, 34% of the reference's own peak. The two hosts converge to different discrete solutions, as expected when they discretize the same continuous law differently: the sweep thus separates a truncation pathology that convergence removes from a discretization difference that it does not.

This cuts two ways: convergence largely cures the amplification, so iterating further or adopting the implicit realization removes it, and the bound of §2 is not a substitute for convergence but targets the regime where neither escape is available. Following the one-iteration small-step regime [12], interactive position-based solvers run schedules near 1×8 to 2×4 , well below the $K \geq 24$ this

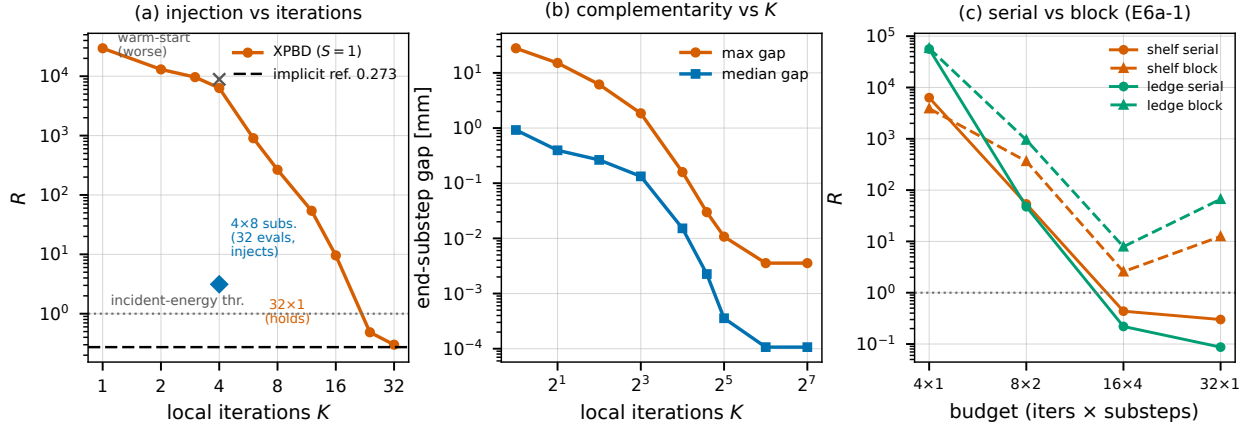


Figure 3: The XPBD operating envelope (our tested position-based host, governor OFF). (a) Injection R vs. local iterations K on a deterministic shelf drop ($S=1$): R decays monotonically toward the implicit reference (0.273), crossing the incident-energy threshold between $K=16$ and 24. Annotated: the 32x1-vs-4x8 equal-row pair (same 32 row evaluations, but spent as substeps it still injects) and warm-starting (no help; worse at 4x1). (b) The end-of-substep gap converges with K , confirming the falling ratio is the row converging, not a scalar shrinking coincidentally. (c) Shared-row treatment (E6a-1): serial per-row support projection (the paper path) vs. joint block condensation, both scenes. The serial path converges/holds by 16x4–32x1; block condensation does *not* — a different, worse convergence path that injects where serial holds. Condensing the shared rows is not the fix.

scene needs. On 18 further cells (3 scenes $\times$ 3 hosts $\times$ $\{1 \times 8, 2 \times 4\}$, relaxation 0.7) the position-based host violates (2) in all six of its deployed cells, worst $R = 2282$ — on 8 row evaluations, twice the 4 of the 4x1 corner — while the others behave as in the sweep (1/6 and 0/6, the former by at most 0.011 J). Governed, all 18 satisfy (2) (worst $R = 1.023$) and the projection is gentler: worst penetration 9.8 mm against 21.6.

Not a knife-edge configuration. Varying one setting at a time away from the two worst cells — h , contact and support compliance, modal rank, both Rayleigh coefficients — (2) is violated with $R > 1$ in all 24 configurations, R spanning 2.5 to 4.7×10^5 . Rank localizes the effect without removing it: excluding the stiff cluster cuts the shelf 4x1 ratio from 6333 to 22.1 (584 J still overdrawn), and across the sweep 6 of 8 injecting cells still overdraw (worst ledge 4x1, $+1.24 \times 10^6$ J), so band-limiting the co-solve reduces but does not remove the injection.

3.3 The cost of enforcement

Accuracy. Bounding the energy is useful only if the bounded solution improves on the unbounded one. At a moderate injecting cell (shelf, 8x2, relaxation 0.7), peak modal energy is 1555.6 J ungoverned and 29.56 J governed against the implicit reference’s 7.92 J: the energy error falls $196\times \rightarrow 3.7\times$ ($+1548 \rightarrow +22$ J), and $189\times \rightarrow 3.6\times$ against the XPBD self-reference (8.22 J), so it does not depend on the reference chosen. The trajectory, however, degrades: $\|d - d^{\text{ref}}\|_\infty$ is 6.5 mm ungoverned and 14.2 mm governed, 33% and 71% of the reference’s peak — the governor is a safety envelope, not an accuracy device. At the deployed 1x8 cell energy gives $2823\times \rightarrow 3.7\times$ and the trajectory improves too ($108 \rightarrow 96\%$ of reference peak): the trade is budget-dependent, but 96% is still not faithful. The 500x1 reference is converged in iteration count,

Table 1: Post-projection contact validity for the *position-based* host at starved budgets, relaxation 0.7, per clamp-active substep. Gap violation is measured immediately after the γ -scale; the corrective impulse is the next substep’s normal impulse against the unclamped steady-state median, a ratio because the hosts’ multipliers are different objects. The 4x1 cells clamp on nearly every substep, so no steady state exists there (—).

scene	budget	clamped	gap viol. med / max [mm]	impulse	λ var.
shelf	4x1	107/108	1.24 / 21.6	—	—
shelf	8x2	171/216	0.21 / 6.43	3.68x	5.6x
ledge	4x1	97/108	0.08 / 21.2	1.03x	1.08x
ledge	8x2	178/216	0.07 / 4.08	8.92x	62x

not schedule-matched to the governed $S>1$ runs, so the residual bounds the governed error rather than isolating truncation from schedule mismatch.

This also explains how a $196\times$ energy error can coexist with a peak deflection only $1.2\times$ too large: the excess is neither kinetic (0.3%) nor low-frequency — the shelf spectrum is bimodal (ten bending modes below 2.1 kHz, six stiff above 20 kHz) and 99.6% of the ungoverned energy sits in the stiff cluster, $\sim 200\times$ the substep rate, against 0.0% for both references. A single scalar cannot redistribute energy across the spectrum: γ corrects the *amount* ($1555.6 \rightarrow 29.3$ J), leaves the character intact ($99.6 \rightarrow 97.7\%$), and scales the legitimate sag down with the noise ($24.1 \rightarrow 5.9$ mm against 20.0); the first-line remedy is band-limiting the co-solved basis to the representable band, not a scalar.

Contact validity. Since $\gamma < 1$ shrinks the sag $U_y^T \mathbf{q}$, (4) lifts the support *into* the resting body without re-solving contact. Table 1

quantifies the consequence, instrumented by wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

Median post-projection penetration is 0.07–1.24 mm, but the worst case reaches 21.6 mm, a 3.1–12.5× increase over its pre-scale value. The shelf board is 30 mm thick, so that is 72% of its thickness; against the unclamped scene at either reference (resting sag 1.7–2.4 mm, peak dynamic deflection 20.0–20.5 mm) it is 9–13× the sag and 1.05–1.08× the board’s peak deflection: a visible interpenetration where the projection is most load-bearing. Contact recovers in the next substep rather than accumulating, but that correction costs a normal impulse of up to 8.9× the steady-state value, with multiplier variance up to 62× higher. Net rigid linear momentum is unchanged across the projection — exactly zero drift, verified numerically — in every clamp-active substep of every cell, necessarily so since (4) writes only modal state.

On the augmented-Lagrangian host the same instrumentation fires only at its one materially-overdrawing cell (the starved 4×1 table corner, +6.7 J), where the projection is an order of magnitude gentler: it opens 1.8 mm against the position-based host’s 21.6, clamps a single substep against its near-continuous firing, and needs no measurable corrective impulse. The hosts differ in the scale of the correction, not its character (full rows in the supplement).

3.4 The reduced response against a full-FEM reference

Bounding the energy is likewise useful only if the bounded response is correct. On the ledge scene we compare the reduced arm against an unreduced FEM solve on the *same* discrete operator (the reduced basis is $\text{eigsh}(\mathbf{K}, \mathbf{M})$ of the operator the reference integrates), so discretization bias cancels. The reduced-to-reference peak-deflection ratio rises monotonically $0.38 \rightarrow 0.54 \rightarrow 0.79 \rightarrow 0.88$ as h refines $1/120 \rightarrow 1/960$, plateauing near 0.9; the residual $\sim 10\%$ is $k=24$ mode truncation on a thick slab, not a coupling error — the ring frequency agrees to 0.4% (78.0 vs 78.3 Hz). The far-field shape agrees too: peak deflection sits at the pedestal antinode, not under the impact, and the falloff tracks the reference at Spearman $\rho = 0.89$ with no fitted attenuation term, in the small-displacement regime.

3.5 Runtime cost

On the CPU host at 16×4, over ten repetitions of 100 frames, the baseline step costs 11.0–125.6 ms by scene and host, and the ledger adds 0.23–0.41 ms — 0.9–3.4% — on the shelf and ledge configurations (on the table its per-repetition spread exceeds its mean, so we bound it only by ~ 1.4 ms; the percentage, not the absolute time, is the portable quantity across machines). The cost is budget-dependent and larger at the deployed schedules: at 1×8 and 2×4 the absolute cost stays comparable (0.30–2.22 ms) but the smaller baseline lifts it to 6.6–34.3%, so enforcement is not free where the budget is tight. We make no unqualified real-time claim; a device-resident *enforced* γ does not exist in this work, so governed device timings lie outside our measurements (§4).

4 Limitations

Stability is not accuracy. Where the projection is load-bearing it opens contact violations up to 21.6 mm and moves the trajectory

away from the implicit reference even as it moves the energy toward it (§3.3): bounded, not faithful, so its honest use is as a safety envelope for budgets that would otherwise diverge, and convergence remains the better remedy wherever affordable. A natural next mechanism scales deviations about the quasi-static equilibrium rather than the whole state, preserving the load-bearing sag; it is unbuilt.

Scope of the row. The coupling is normal-only — friction stays body-local, so scraping and rolling transfer lie outside it — and the position-level form fixes $e = 0$. The bound governs modal storage only, not energy created spuriously in rigid degrees of freedom: in the impulse realization, box-box rows rectifying a support ring into net rigid motion, to which the ledger is blind by design.

The supply is a gross sum, so it can be recycled. Energy returning from the modes to the rigid bodies and dissipated there is credited twice. We bound the exposure by accumulating the opposite channel, $\sum_k \max(-\Delta E_{\text{rig}}^k, 0)$, as a fraction of the supply: 0.4–27% (implicit), 3–32% (position-based) — larger than the $\leq 1\%$ we expected — and 102–118% on the augmented-Lagrangian host, whose rigid subsystem gains more than it loses in every cell, the box-box rectification above rather than a property of the bound. Tightening the envelope needs the signed or per-interface accounting this scalar reservoir avoids. The supply also depends on the substep *partition*: a finer schedule rectifies more fluctuation into supply, so the budget belongs to the schedule as well as the scene.

Coverage. Enforcement is host-side and single-machine (CPU float64); a device-resident governor and a validation of the *governed* path against a full-FEM reference remain future work. Three formulations, one implementation of each; three scenes, one contact regime; the augmented-Lagrangian host still overdraws in 3 of 24 cells (up to 6.7 J), so any claim that descent-class solvers are passive in this setting is an empirical statement about this sweep, not a theorem.

5 Conclusion

For a developer who has already chosen a fixed-budget XPBD rigid-body host and wants low-cost vibration on a stiff prop, the decision is not XPBD versus another solver but an ordered one. First pick the representation: a full-space XPBD deformable for large, local or strongly nonlinear deformation, a compact modal state for stiff, small-displacement ringing. If the contact architecture is still open, the tested stiffness-aware implicit realization is the cleanest control we observed — it never overdraws — though that recommendation is implementation-specific, not a solver-class theorem. If the XPBD host is fixed, do not tune blindly: the direct two-way modal row overdraws its measured supply by up to 4.4×10^7 J at a few local iterations, and neither more substeps at equal row cost, nor warm-starting, nor condensing the shared support rows into one block removes it (§3.2); converging the row with iterations does, so spend the local budget there and verify the modal basis. Only when a fixed host still cannot converge does the cumulative modal-storage bound earn its place — it holds in every one of the 90 measured cells, but opens up to 21.6 mm of penetration, so it is a containment envelope, not corrected contact. A low-dimensional modal extension is cheap in state size; it is not automatically easy for a truncated local contact solve.

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